Template Week 2 – Logic

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Assignment 2.1: Parking lot

Which gates do you need?

- To determine whether a parking-lot lamp should turn red, you only need the logical AND operator. The AND operator is used to check whether all parking spaces are occupied at the same time. If every spot is taken at the same time, the lamp will turn red.
  - RED = A AND B AND C

  Even if one spot is empty, the lot is not full, so the lamp is not on.

Complete this table.

Parking lot 1	Parking lot 2	Parking lot 3	Result (full)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Assignment 2.2: Android or iPhone

Which gates do you need?

- When choosing between an iPhone and an Android, you use the XOR operator because exactly one of the two to be true for a choice to be made. If you choose iPhone or choose Android, then you have selected a phone.

Complete this table

Android phone	iPhone	Result (Phone in possession)
0	0	0
0	1	1 (Iphone)
1	0	1 (Android)
1	1	0

Assignment 2.3: Four NAND gates

Complete this table

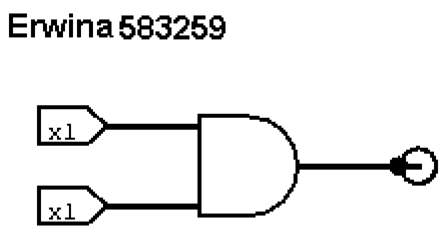
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

How can the design be simplified?

- What the circuit is doing is turning on the output Q, when A and B are different, and it turns off when they are the same. So, instead of using this circuit, you could instead simplify it to a XOR gate.

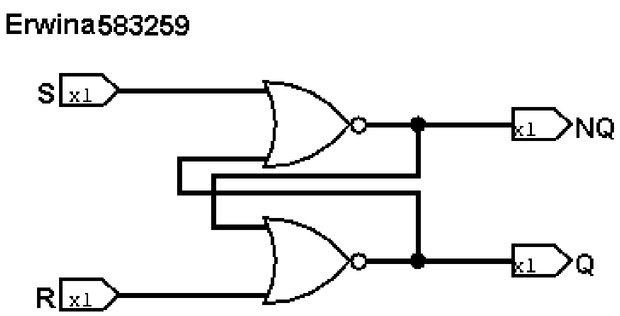
Assignment 2.4: Getting to know Logisim evolution

Screenshot of the design with your name and student number in it:



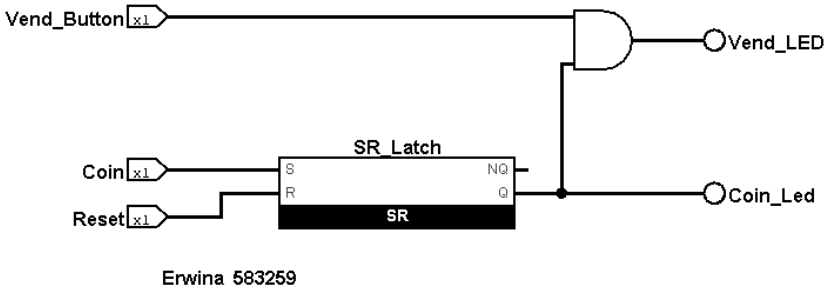
Assignment 2.5: SR Latch

Screenshot SR Latch in Logisim with your name and student number:



Assignment 2.6: Vending Machine

Screenshot Vending Machine in Logisim with your name and student number:



Assignment 2.7: Bitwise operators

Complete the java source code for bitwise operators. Put the source code here.

#1 even or odd

```
int number = 4;
if ((number & 1) == 1) {
    SaxionApp.printLine("number is odd");
} else {
    SaxionApp.printLine("number is even");
}
```

#2 power of 2

```
int number = 5;

if ((number & (number - 1)) == 0 && number > 0) {
    SaxionApp.printLine(number + " is a power of 2");
} else {
    SaxionApp.printLine(number + " is NOT a power of 2");
}
```

#3 check permissions

```
final int READ = 4;
final int WRITE = 2;
final int EXECUTE = 1;

int userPermissions = 7;

if ((userPermissions & READ) != 0) {
    SaxionApp.printLine("user has read permissions");
} else {
    SaxionApp.printLine("user can't read. No permission.");

}
```

#4 assign permissions

```
final int READ = 4;
final int WRITE = 2;
```

```
final int EXECUTE = 1;

    int userPermissions = 0;

    userPermissions = userPermissions | READ;
    userPermissions = userPermissions | EXECUTE;

    if ((userPermissions & READ) != 0) {
        SaxionApp.printLine("Can read");
    } else {
        SaxionApp.printLine("Cannot read");
    }

    if ((userPermissions & WRITE) != 0) {
        SaxionApp.printLine("Can write");
    } else {
        SaxionApp.printLine("Cannot write");
    }

    if ((userPermissions & EXECUTE) != 0) {
        SaxionApp.printLine("Can execute");
    } else {
        SaxionApp.printLine("Cannot execute");
    }
```

#5 update permission

```
final int READ = 4;
final int WRITE = 2;
final int EXECUTE = 1;

int userPermissions = 6;
userPermissions = userPermissions ^ WRITE;
SaxionApp.printLine("User permissions: " + userPermissions);
```

#6 Two's compliment

```
int number = 5;

int negative = ~number + 1;
int backToPositive = ~negative + 1;

SaxionApp.printLine("Original number: " + number);
SaxionApp.printLine("Negative number: " + negative);
SaxionApp.printLine("Back to positive: " + backToPositive);
```

Assignment 2.8: Java Application Bit Calculations

Create a java program that accepts user input and presents a menu with options.

1. Is number odd?
2. Is number a power of 2?
3. Two's complement of number?

Implement the methods by using the bitwise operators you have just learned.

Organize your source code in a readable manner with the use of control flow and methods.

Keep this application because you need to expand it in week 6 for calculating network segments.

Paste source code here, with a screenshot of a working application.

```
public static void main(String[] args) {
       SaxionApp.start(new Application(), 800, 800);
    }

    public void run() {
        SaxionApp.print("Enter a number: ");
        int userInput = SaxionApp.readInt();

        boolean programIsRunning = true;
        while (programIsRunning){
            SaxionApp.clear();

            SaxionApp.printLine("--- MENU ---");
            SaxionApp.printLine("Selected number: " + userInput, Color.ORANGE);
            SaxionApp.printLine("1. Odd or even.");
            SaxionApp.printLine("2. Power of 2");
            SaxionApp.printLine("3. Two's complement");
            SaxionApp.printLine("0. Exit");
            SaxionApp.print("Choose an option: ");

            int userChoice = SaxionApp.readInt();

            switch (userChoice) {
                case 1:
                    if (isOdd(userInput)) {
                        SaxionApp.printLine(userInput + " is odd.");
                    } else {
                        SaxionApp.printLine(userInput + " is even.");
                    }
                    break;

                case 2:
                    if (isPowerOfTwo(userInput)) {
                        SaxionApp.printLine(userInput + " is power of 2.");
                    } else {
                        SaxionApp.printLine(userInput + " is NOT power of 2.");
                    }
                    break;

                case 3:
                    int negative = twosComplement(userInput);
                    SaxionApp.printLine(userInput + " Two's complement of " +
userInput + " is: " + negative);
                    break;
                case 0:
```

```java
                    programIsRunning = false;
                    SaxionApp.printLine("Goodbye!!");
                    continue;

                default:
                    SaxionApp.printLine("Something went wrong :/");
            }

            SaxionApp.pause();

        }



    }

    public static boolean isOdd(int number) {
        return ((number & 1) == 1);
    }

    public static boolean isPowerOfTwo(int number) {
        if (number <= 0) return false;
        return (number & (number - 1)) == 0;
    }

    public static int twosComplement(int number) {
        return ~number + 1;
    }


}
```